Year 8
Mathematics

Unit 2: Equations and Formulae

2.3 More Complex Equations

More Complex Equations

STARTER · 5 MIN

Date: _____ **Title: More Complex Equations**

LEARNING OBJECTIVES

- To be able to write and solve equations with letters on both sides

STARTER *Answer in your book. No calculators.*

1. Solve $2x + 3 = 11$
2. Solve $5x - 1 = 24$
3. Simplify $4x - 2x$
4. Simplify $7x - 3x + 2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $x = 4$

2. $x = 5$

3. $2x$

4. $4x + 2$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use inverse operations to solve equations with letters on both sides	☐
SILVER	I can check the solution by substituting back into the original equation	☐
GOLD	I can write a complex equation from a word problem and solve it correctly	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Complex Equations

When the letter is on both sides, remove the smallest letter term first.

Letters both sides

e.g. $4x + 7 = 2x + 15$

Subtract $2x$ from both sides

Then two steps

$2x + 7 = 15$

$2x = 8$, $x = 4$

Check

Both sides must match.

$4 \times 4 + 7 = 23$ and $2 \times 4 + 15 = 23$

No solution

The letters cancel and it is false.

$2x + 9 = 2x + 3 \rightarrow 9 = 3$: false

Examples 1 and 2 — Letters on Both Sides

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Remove the smallest x term Then solve in two steps

Example 1

Solve

$4x + 7 = 2x + 15$

Example 2

Solve

$5x - 3 = 3x + 7$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Subtract 2*x* from both sides

$$2x + 7 = 15$$
$$2x = 8$$
$$\boldsymbol{x = 4}$$

Tip: remove the smaller *x* term.

Example 2

Subtract 3*x* from both sides

$$2x - 3 = 7$$
$$2x = 10$$
$$\boldsymbol{x = 5}$$

Tip: check: 22 = 22.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve*

Hint Remove the smallest x term Then solve in two steps

1. $5x + 2 = 3x + 10$

2. $6x + 1 = 2x + 9$

3. $7y - 4 = 4y + 8$

4. $3m + 11 = m + 21$

5. $8a - 5 = 5a + 1$

★ $9n - 7 = 4n + 13$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $x = 4$
2. $x = 2$
3. $y = 4$
4. $m = 5$
5. $a = 2$

★ $n = 4$

Examples 3 and 4 — Checking and Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Expand brackets first Substitute to check

Example 3

Solve and check

$3(x + 2) = x + 14$

Example 4

Solve

$2x + 9 = 2x + 3$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Expand, then remove *x*

$3x + 6 = x + 14$

$2x + 6 = 14$

$2x = 8$

$\boldsymbol{x = 4}$

Tip: check: 3 × 6 = 18 = 4 + 14.

Example 4

Subtract $2x$ from both sides

$9 = 3$

This is false, so there is

no solution.

Tip: not every equation can be solved.

Task 2

PRACTICE · 9 MIN

SILVER *Solve and check*

Hint Expand brackets first Substitute to check

1. $2(x + 4) = x + 11$

2. $4(y - 1) = 2y + 6$

3. $5(a + 1) = 3(a + 3)$

4. $3x + 5 = 3x + 8$

5. $10 - x = 2x + 1$

★ $6(m - 2) = 4m$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x = 3$
2. $y = 5$
3. $a = 2$
4. No solution
5. $x = 3$

★ $m = 6$

Examples 5 and 6 — Equations from Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Equal costs: equate them Equal sides: equate them

Example 5

(2 marks)

Ana has $x + 12$ sweets.

Ben has $3x$ sweets.

They have the same number.

Find x.

Example 6

(3 marks)

Both rectangles have the same area. A is x cm by 4 cm, B is $(x - 2)$ cm by 6 cm.

Find x.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Same number: make them equal

$3x = x + 12$

$2x = 12$

$\boldsymbol{x = 6}$

Tip: each has 18 sweets.

Example 6

Write each area, then make them equal

$4x = 6(x - 2)$

$4x = 6x - 12$

$12 = 2x$

$\boldsymbol{x = 6}$

Tip: both areas are 24 cm^2.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: write and solve*

Hint Equal costs: equate them Equal sides: equate them

1. $5x$ and $2x + 18$ are equal. Find x.

2. Plan A: \$30 + \$5 a day. Plan B: \$10 a day. Same when?

3. Think of a number: × 5 − 3 gives the same as × 3 + 7. Number?

4. Square sides $2x + 1$ and $x + 6$. Find x.

5. $4(x - 1) = 2(x + 5)$

★ Does $3(x + 2) = 3x + 5$ have a solution?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $x = 6$
2. $30 + 5d = 10d$: 6 days
3. $5n - 3 = 3n + 7$: $n = 5$
4. $x = 5$
5. $x = 7$

★ No: $6 \neq 5$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $4x + 7 = 2x + 15$

SILVER Solve $5x - 3 = 3x + 7$

GOLD Solve $2x + 9 = 2x + 3$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = 4$

SILVER $x = 5$

GOLD **No solution**

Exam Practice — Letters on Both Sides

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Remove the smallest x term
Then solve in two steps

Question 1

(2 marks)

Solve $6x + 3 = 2x + 19$

$x =$ .....

Question 2

(2 marks)

Solve $7y - 5 = 3y + 11$

$y =$ .....

Exam Practice — Brackets

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Expand brackets first Substitute to check

Question 3

(3 marks)

Solve $3(x + 4) = x + 20$

$x =$ .....

Question 4

(2 marks)

Kai says $2x + 5 = 2x + 1$ has the solution $x = 2$.

Explain why he is wrong.

.....

Exam Practice — Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
Equal costs: equate them
Equal sides: equate them

Question 5

(3 marks)

Plan A costs \$20 plus \$3 per GB.
Plan B costs \$8 plus \$5 per GB.

For how many GB do both plans cost the same?

..... GB

Question 6

(3 marks)

The sides of an equilateral triangle are $3x - 1$, $x + 9$ and $2x + 4$.

Find x and the perimeter.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$4x + 3 = 19$ — M1

$\mathbf{x = 4}$ — A1

Tip: subtract $2x$ first.

Question 2

$4y - 5 = 11$ — M1

$\mathbf{y = 4}$ — A1

Tip: subtract $3y$ first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$3x + 12$	$= x + 20$	M1
$2x$	$= 8$	M1
$\boldsymbol{x}$	$\boldsymbol{= 4}$	A1

Tip: expand the bracket first.

Question 4

Subtracting $2x$ gives $5 = 1$	M1
False: there is no solution	A1

Tip: check $x = 2$: $9 \neq 5$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$20 + 3g = 8 + 5g$ — M1

$12 = 2g$ — M1

$g = 6$ — A1

Tip: check: both cost $38.

Question 6

$3x - 1 = x + 9$ — M1

$x = 5$ — A1

Perimeter $3 \times 14 = 42$ — A1

Tip: all three sides are equal.